EXAMPLE 10

Three-Story Plane Frame with ADAS Elements - Nonlinear Time History Analysis

Problem Description

This is a single bay three-story plane frame subjected to ground motion, as shown in Figure 10-1. The El Centro 1940 (N-S) record is used in the nonlinear time history analysis. Three elements that absorb energy through hysteresis (ADAS elements as described in Scholl 1993 and Tsai, et al. 1993) are used to connect the chevron braces to the frame. Two models are investigated. In the first model, the ADAS elements are intended to produce about 5% damping in the fundamental mode. In the second model, damping is increased to 25%. The manufacturer supplied the properties of the ADAS elements.

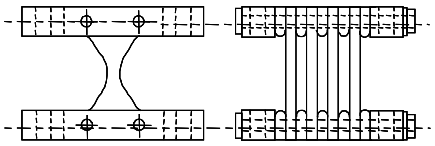
The ADAS elements are modeled in ETABS by assigning a panel zone with a nonlinear link property to the mid-span point object where the chevrons intersect the beams at each story. The link properties use the uniaxial hysteretic spring property (PLASTIC1) and provide beam-brace connectivity with nonlinear behavior in the U2 (shear in the 1-2 plane) direction. Under this arrangement, displacements are transferred between the chevrons and the frame via the link elements undergoing shear deformation.

Geometry, Properties and Loads

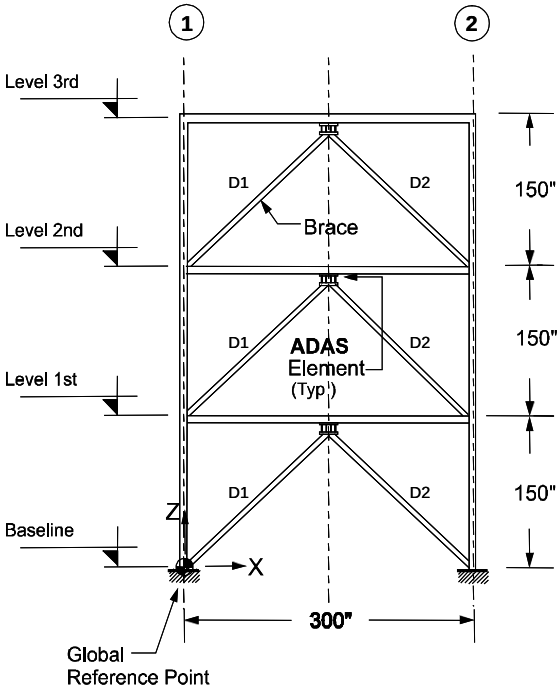
The frame is modeled as a two-column line, one-bay system. Kip-inch-second units are used. The modulus of elasticity is taken as 29000 ksi. Column, beam and brace section properties are user-defined.

A single rigid diaphragm is allocated to each story level and connects all three point objects (two column points and one mid-span point) at each story. Because of the rigid diaphragms, no axial force will occur in the beam members. All members are assigned a rigid zone factor of 1.

In both models the value of post yield stiffness ratio is taken as 5% and the time increment for output sampling is specified as 0.02 second.



Typical ADAS Element



Frame Elevation

Figure 10-1 Planar Frame with ADAS Elements

Technical Features of ETABS Tested

- Two-dimensional frame analysis
- Panel zones
- Point assignments
- Nonlinear time history analysis
- Ritz vectors

Results Comparison

Sample results are compared in Table 10-1 with results from the nonlinear analysis program DRAIN-2DX (Prakash, et al. 1993) for both 5% and 25% damping cases.

Table 10-1 Results Comparison

	5% Damping		25% Damping	
Level	**ETABS**	**DRAIN-2DX**	**ETABS**	**DRAIN-2DX**
Comparison of Maximum Story Deflections				
3rd	4.57	4.57	2.10	1.92
2nd	3.48	3.51	1.68	1.55
1st	1.82	1.82	0.92	0.86
Comparison of Maximum Link Shear Force				
3rd	7.29	7.31	17.75	17.40
2nd	13.97	13.92	36.70	36.20
1st	17.98	18.00	47.79	47.10
Comparison of Maximum Brace Axial Force				
3rd	5.16	5.17	12.55	12.30
2nd	9.88	9.84	25.95	25.60
1st	12.71	12.70	33.79	33.28

Computer Files

The input data files for this example are Example 10A.EDB (5% damping) and Example 10B.EDB (25% damping). The time history file is ELCN-THE. These files are provided as part of the ETABS installation.

Conclusions

The results comparison show acceptable to exact agreement between ETABS and DRAIN-2DX.